

# Designing Self-Aware Multi-Agent AI Systems: A Two-Fold Framework Based on AIBOM and Reflective Architecture



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- Increasing complexity in LLM-based AI systems
  - Drives the adoption of **multi-agent Cognitive Workflows**, where specialized AI agents collaborate
  - Leads to a growing need for **Reliability**, **Maintainability**, and **Flexibility** [1] in system behaviour
- Multi-agent AI systems
  - Require coordination among diverse agents
  - Depend on a variety of tools and data sources to operate properly
- Regulatory Pressure: emerging regulations (e.g., **EU AI Act**) emphasize transparency and accountability
- Unresolved gaps in literature:
  - Lack of system-level design patterns for multi-agent workflows
  - Absence of Comprehensive **AI Traceability**: existing tools (e.g., Model/Dataset Cards, DataBOM [2,3]) provide only partial traceability, inadequate for complex systems

[1] ISO/IEC 25010:2011. Systems and Software Engineering – Quality Requirements and Evaluation (SQuaRE).

[2] Barday, I., Preece, A., Taylor, I., & Verma, D. (2019). Towards traceability in data ecosystems using a bill of materials model.

[3] Mitchell, M., Wu, S., Zaldivar, A., et al. (2019). Model cards for model reporting. In Proc. FAT, pp. 220–229. ACM.

## Self-awareness and runtime adaptability

- Evolution of the **Reflection Architectural Pattern** [1] applied to agent systems
- Dynamic management of **Meta-models Repository**
- **Semantic interoperability**: heterogeneous components with no prior knowledge of each other can interact seamlessly in a cohesive workflow
- Builds upon SALLMA architecture [2]



## Traceability

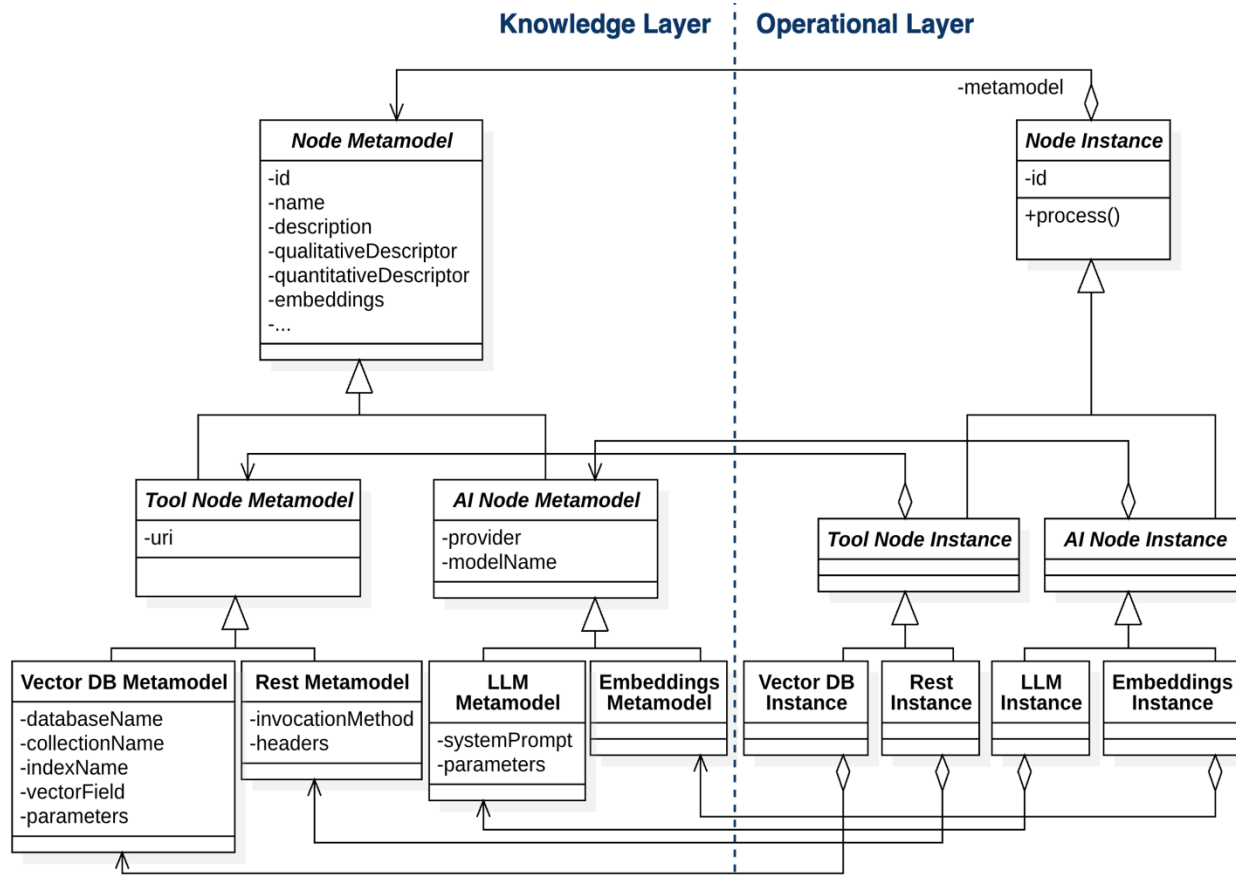
- Introduces **Artificial Intelligence Bill of Materials** (AIBOM) formalism
- Nascent concept evolving from the Software Bill of Materials (**SBOM**)
- Extends supply chain management techniques to AI systems (e.g., models, datasets, prompts, and tools)



**Core contribution:** the integration of the Reflection Pattern (for structure/adaptability) and the AIBOM (for traceability) represents the *novel and main* contribution of this work, offering a response to the identified gaps

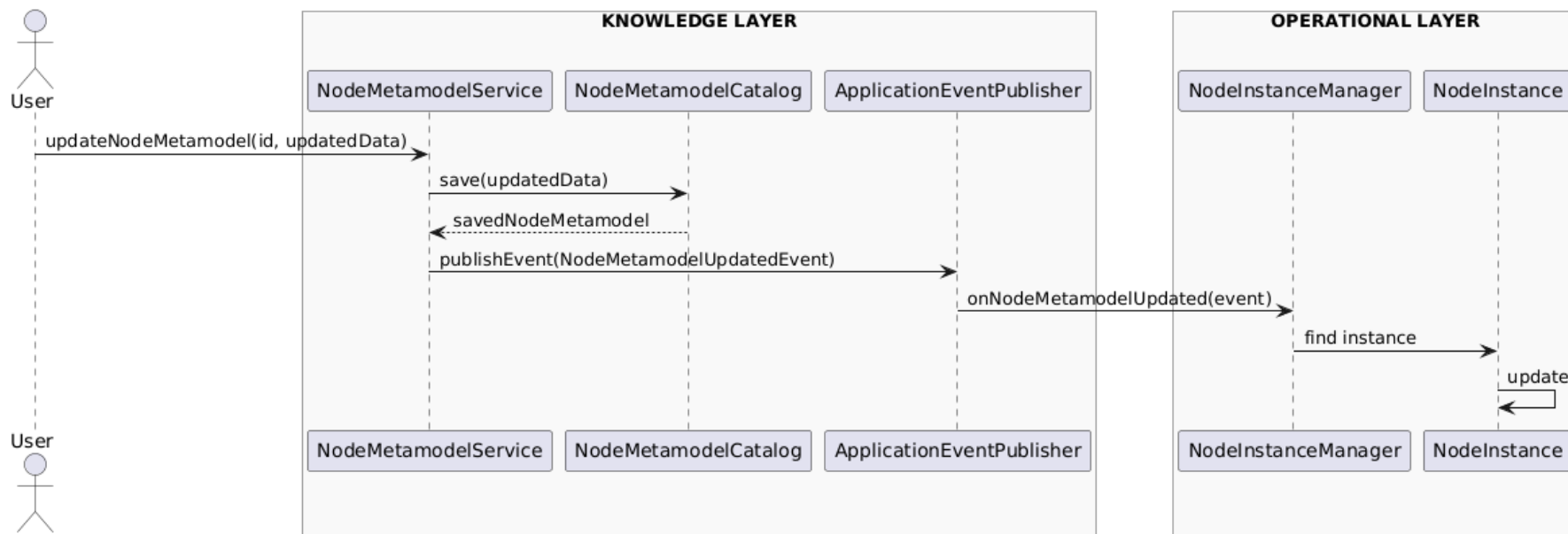
[1] Frank Buschmann, Regine Meunier, Hans Rohnert, Peter Sommerlad, and Michael Stal. Pattern-Oriented Software Architecture: A System of Patterns. John Wiley & Sons, Chichester, UK, 1996.

[2] Marco Becattini, Roberto Verdecchia, and Enrico Vicario. Sallma: A software architecture for llm-based multi-agent systems. In SA Trends 2025. Software Technologies Laboratory, University of Florence, 2025.

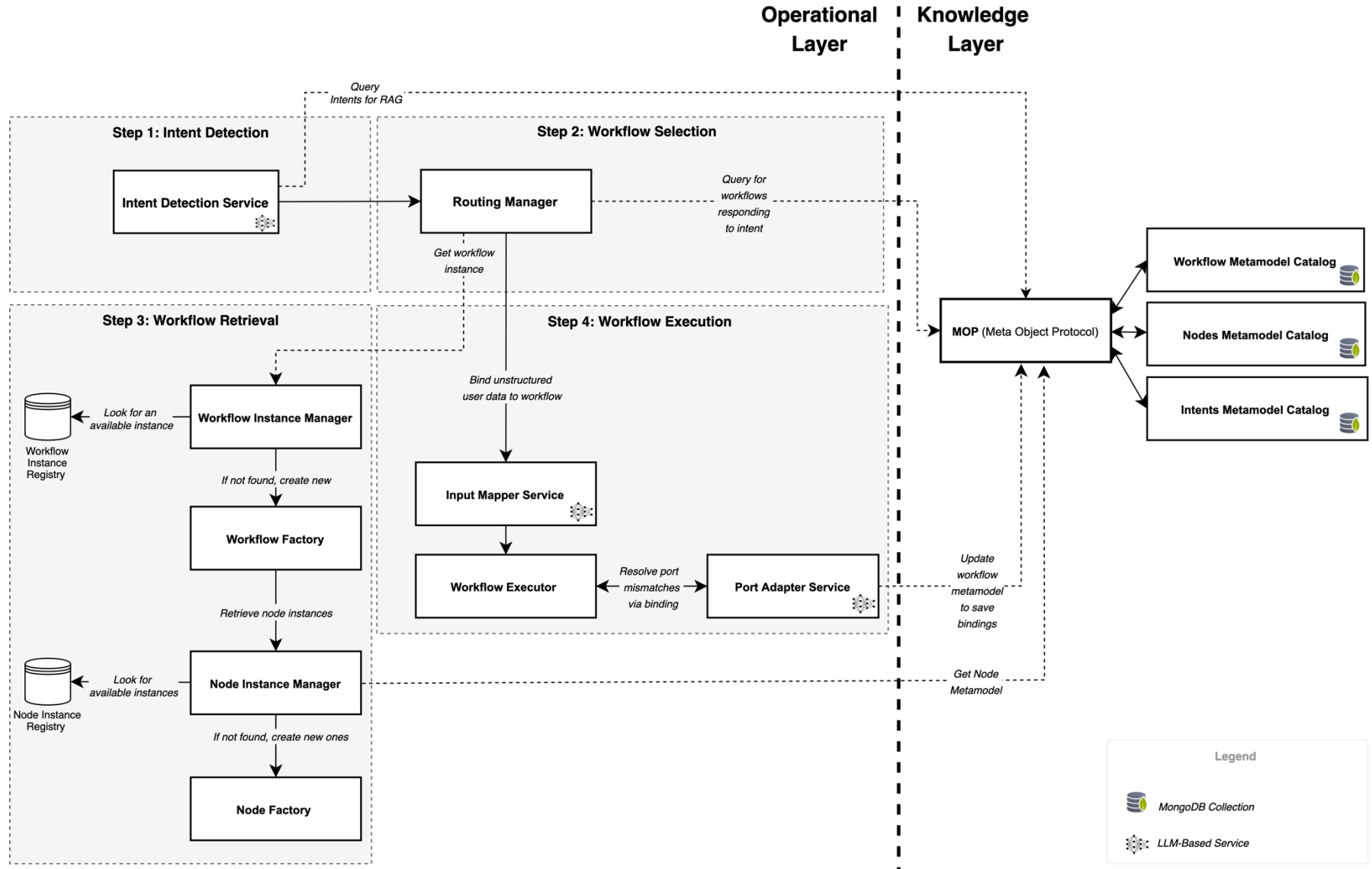


- The architecture follows a dual-layer structure, consisting of a Knowledge Layer and an Operational Layer:
  - The **Knowledge Layer** is responsible for managing workflow and agent configuration metamodels
  - The **Operational Layer** handles real-time task execution and agent interactions
- Within both layers, specialized nodes represent diverse AI agents and tools (e.g., LLMs, Embedding Models, HTTP-REST Tools, Vector Databases)

# System Architecture: Runtime Adaptations



- Operational agents interact with the Knowledge Layer to retrieve information
- Agents update knowledge at **runtime** to reflect environmental changes or new user requirements
- Knowledge Layer updates may originate from:
  - External sources (e.g., new software releases, dependency upgrades)
  - Internal triggers by operational agents (e.g., the Port Adaptor Service)
- Metamodel updates are propagated via **events**; subscribed instances update their configurations



- **Java 17**: Used as the core implementation language for the entire architecture
- **Spring AI** [1]
  - Selected over **LangChain**
  - Enabled structured interaction with foundation models, embedding models, and external tools
  - Supported function calling, structured output parsing, and prompt augmentation
- **MongoDB**
  - Served as the meta-model repository, storing nodes, workflows, and AIBOM entries
  - Atlas **Hybrid Search** combined full-text and vector similarity search, enabling:
    - Semantic retrieval of components and intents
    - Dynamic selection of nodes based on LLM-interpreted requirements using embedded descriptors
- Model Providers:
  - Anthropic – Claude 3.5 Sonnet (For Input Mapping and Port Adaption)
  - OpenAI – GPT-4o (For Intent Detection)
  - OpenAI – text-embedding-3-small (For generating embeddings)

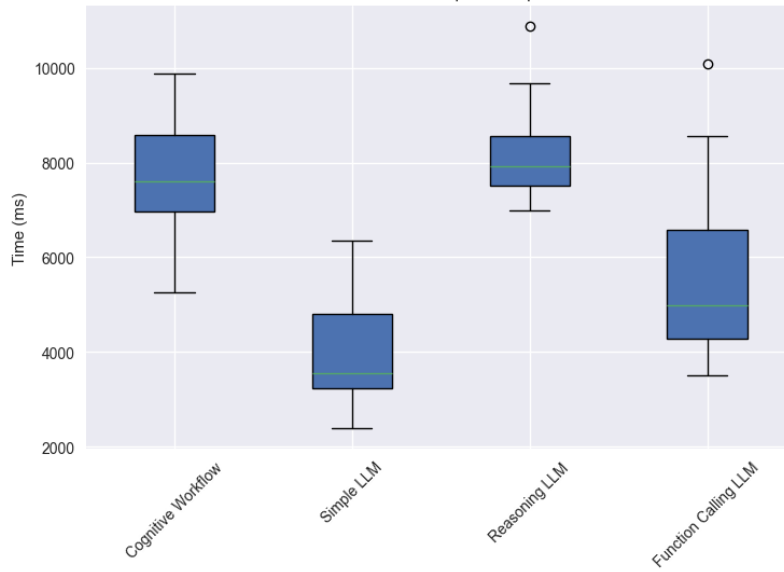
[1] Spring AI: An Application Framework for AI Engineering. (2025). The Spring Team. <https://spring.io/projects/spring-ai>

- *Scenario:* An intelligent system routes user tasks across distributed, heterogeneous cloud and network resources based on user intent and dynamic constraints (e.g., latency, energy efficiency, cost)
  - **AI for Networking** (AI4NE): employing AI models to support routing decisions
  - **Networking for AI** (NE4AI): use of programmable networks to meet AI workload demands
- *Goal:* Demonstrate the relevance of the Reflection pattern and Cognitive Workflows in enabling semantic, cross-vendor, and intent-driven workflows
- Each node in the workflow has characteristics that are not known in advance. Nodes can be LLM-based, but also non-AI-based
- **Cross-Vendor Compatibility:** Requests are expressed in natural or unstructured language, enabling interoperability across various service providers and removing the need for predefined formatting
- Workflow pipeline:
  - 1. Intent Detection and Requirement Extraction** (LLM)
  - 2. Hardware Retrieval** (Tool)
  - 3. Hardware Selection** (LLM)
  - 4. Routing Computation** (Tool)
  - 5. Route Finalization** (LLM)

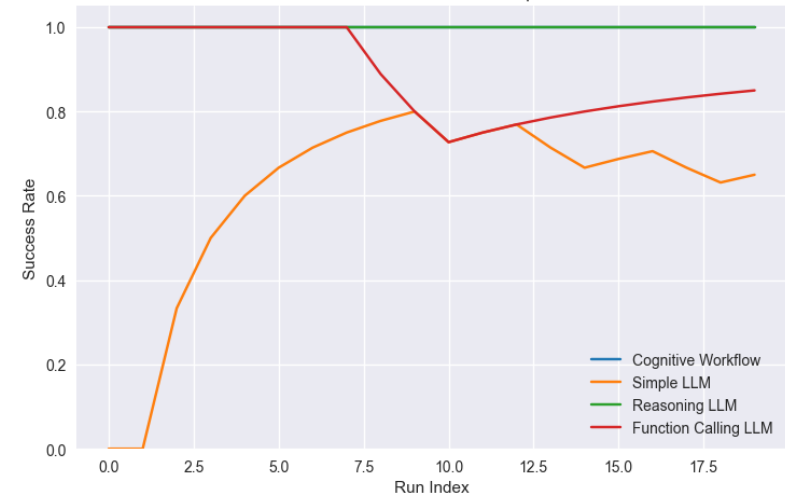
# Use case: Experimental Results

1

Execution Time Boxplot Comparison

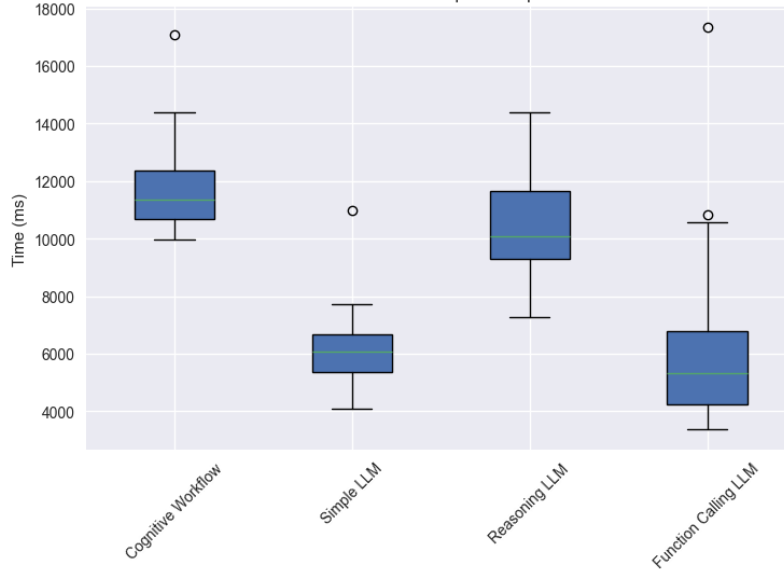


Cumulative Success Rate Comparison

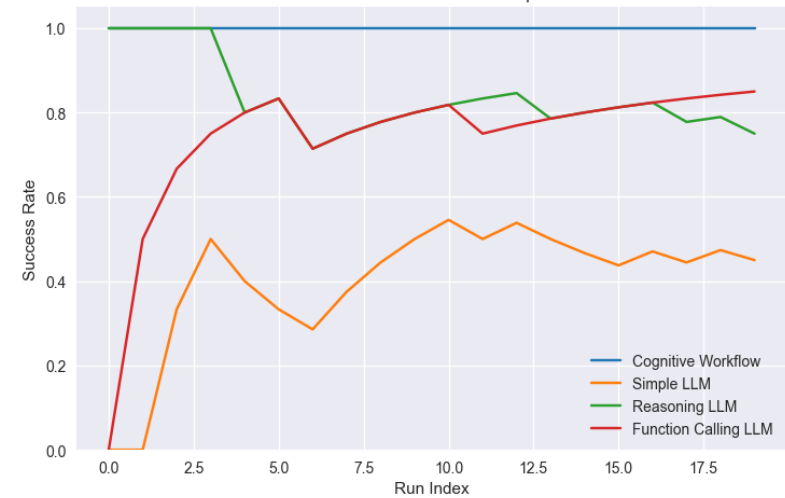


2

Execution Time Boxplot Comparison



Cumulative Success Rate Comparison



- This thesis lays the groundwork for more trustworthy and adaptable AI systems by:
  - Bridging key gaps in AI system-level design and traceability
  - Aligning with future regulatory requirements
- The study presents a comprehensive and well-structured prototype that:
  - Demonstrates core concepts and methodologies
  - Validates the feasibility of the proposed approach
  - Provides a structured foundation for ongoing research
- This work opens the way to further research, including:
  - **Quantitative evaluations**
    - e.g., the analysis of the effects deriving from the modification of the SBOM
  - **Knowledge Subtraction scenarios**
    - e.g., the impact assessment of knowledge modification on system outputs

For reproducibility: thesis, source code, data, and experimental artifacts accessible via doi at <https://doi.org/10.5281/zenodo.15774796>

# Thank you for your time and attention

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